

AI-BASED RESUME SCREENING SYSTEM

MACHINE LEARNING & EXCEL DASHBOARD
PROJECT

PROJECT REPORT

BY

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1. INTRODUCTION

Recruitment is one of the most important processes in any organization. Companies often receive a large number of resumes for a single job opening, making manual screening difficult, time-consuming, and inconsistent. Recruiters must review candidate details such as skills, experience, education, certifications, projects, salary expectation, and overall suitability for the role.

Traditional resume screening methods may overlook important candidate information and can delay the hiring process. To overcome this challenge, this project develops an AI-Based Resume Screening System using machine learning techniques to predict whether a candidate is likely to be selected or rejected based on resume-related attributes.

Why Resume Screening Matters

- For Recruiters: Reduces manual effort and speeds up candidate shortlisting.
- For Organizations: Helps make recruitment faster, more structured, and data-driven.
- For Candidates: Supports fairer and more consistent evaluation.
- For Analytics: Helps identify the key factors influencing hiring decisions.

The objective of this project is to use machine learning algorithms to automate resume screening, compare model performance, and deploy the best-performing model through Streamlit for real-time candidate prediction. In addition, an Excel dashboard was created to summarize candidate selection trends and model insights.

2. ABSTRACT

This project focuses on building an AI-Based Resume Screening System using multiple machine learning classification algorithms. The dataset contains candidate-related attributes such as skills, experience, education, certifications, job role, salary expectation, projects count, and AI score, along with recruiter decision labels.

The project involved data preprocessing, exploratory data analysis (EDA), model training, model comparison, and hyperparameter tuning. Several classification models were trained and evaluated, including Logistic Regression, Decision Tree, and Random Forest. Hyperparameter tuning was performed using GridSearchCV to improve model performance and generalization.

The results show that Random Forest and Decision Tree achieved very high accuracy, while Logistic Regression provided a useful baseline model. To make the project more practical, the final model was deployed using Streamlit, allowing users to enter candidate details and instantly receive a prediction as Selected or Rejected. Additionally, the project results were exported to Excel and used to create an interactive dashboard for candidate analysis and selection trends.

3. OBJECTIVES

- Analyse resume-related candidate data to understand hiring patterns.
- Perform Exploratory Data Analysis (EDA) to study candidate attributes and decision trends.
- Build and evaluate multiple machine learning classification models for resume screening.
- Perform hyperparameter tuning to improve model performance.
- Identify the most important features influencing candidate selection.
- Export results to Excel and create a dashboard for analytical reporting.
- Deploy the final model using Streamlit for real-time prediction.

4. METHODOLOGY

Step 1: Data Collection

- Dataset collected for the AI-Based Resume Screening System project.
- The dataset contains candidate-related attributes and recruiter decisions.

Step 2: Data Preprocessing

- Loaded the dataset into Python using Pandas.
- Checked the dataset structure, column names, and data types.
- Verified missing values and ensured the dataset was clean.
- Separated input features and target variable.
- Prepared the dataset for model training and testing.

Step 3: Exploratory Data Analysis (EDA)

- Analysed candidate distribution across selection and rejection classes.
- Studied relationships between AI Score, Experience, Skills, Salary, and Recruiter Decision.
- Visualized feature distributions and trends using graphs.

Step 4: Feature Selection and Train-Test Split

- Selected relevant input features such as:
 - Skills
 - Experience (Years)
 - Education
 - Certifications
 - Job Role
 - Salary Expectation
 - Projects Count
 - AI Score

- Step Target variable used: Recruiter Decision / Predicted Decision
- Split the dataset into training and testing sets.

Step 5: Model Building

Applied multiple machine learning classification algorithms:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Step 6: Hyperparameter Tuning

- Performed Evaluation tuning using GridSearchCV.
- Tuned selected models such as:
 - Logistic Regression
 - Random Forest
- Compared the best cross-validation scores obtained after tuning.

Step 7: Model Evaluation

- Evaluated models using accuracy and comparison of results.
- Compared performance of all trained models.

Step 8: Feature Importance Analysis

- Analysed the most influential features affecting candidate selection.

Step 9: Result Export and Dashboard

- Exported prediction results to Excel.
- Created an Excel dashboard to visualize candidate selection statistics and model insights.

Step 10: Streamlit Deployment

- Deployed the final model using Streamlit.
- Built a simple web app for real-time candidate screening.

5. DATASET OVERVIEW

Project Title: AI-Based Resume Screening System

- Dataset Type: Candidate screening / recruiter decision dataset
- Total Records: 200 candidate records
- Target Variable: Predicted Decision / Actual Decision

Features Used

- Skills
- Experience (Years)
- Education
- Certifications
- Job Role
- Salary Expectation (\$)
- Projects Count
- AI Score (0–100)

Target Output

- Selected
- Rejected

The dataset is designed to simulate a resume screening process where candidate attributes are used to predict recruiter decisions.

6. DATA ANALYSIS & EDA VISUALS

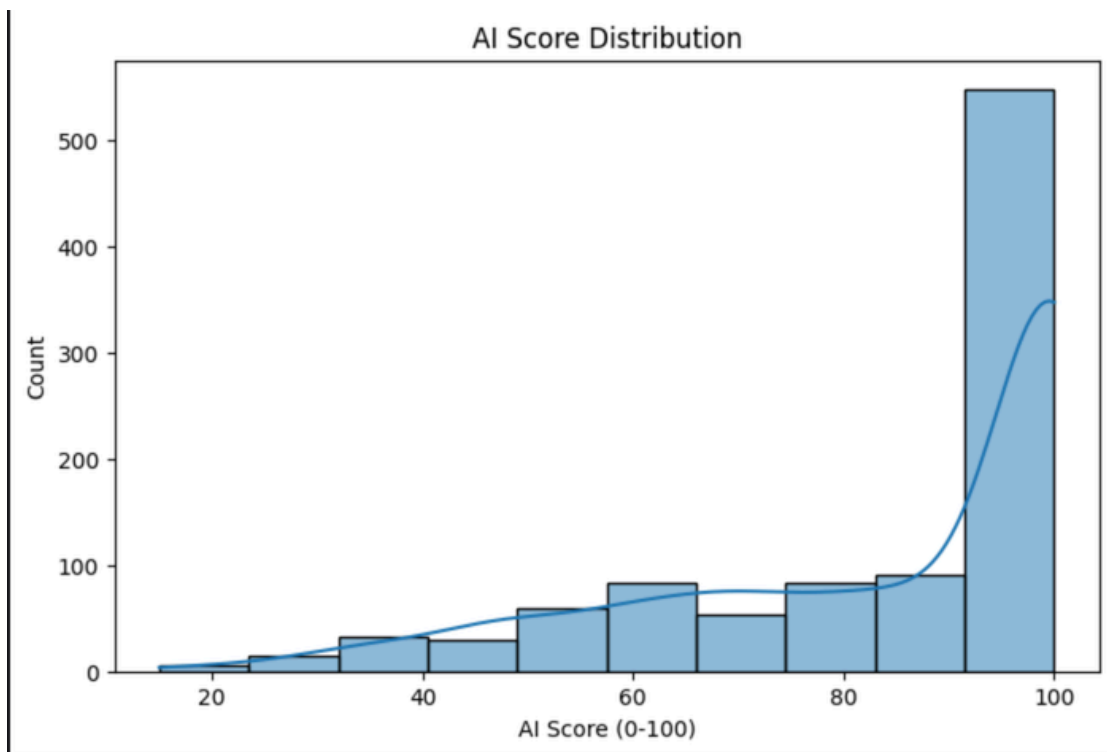
Key Observations

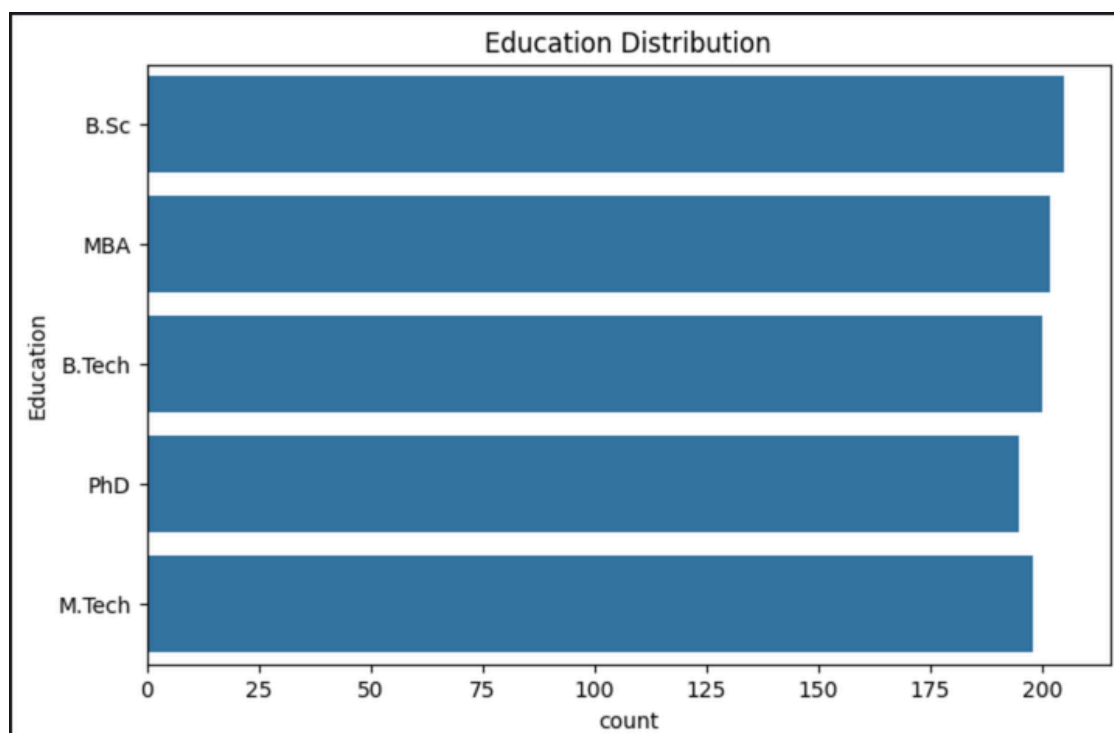
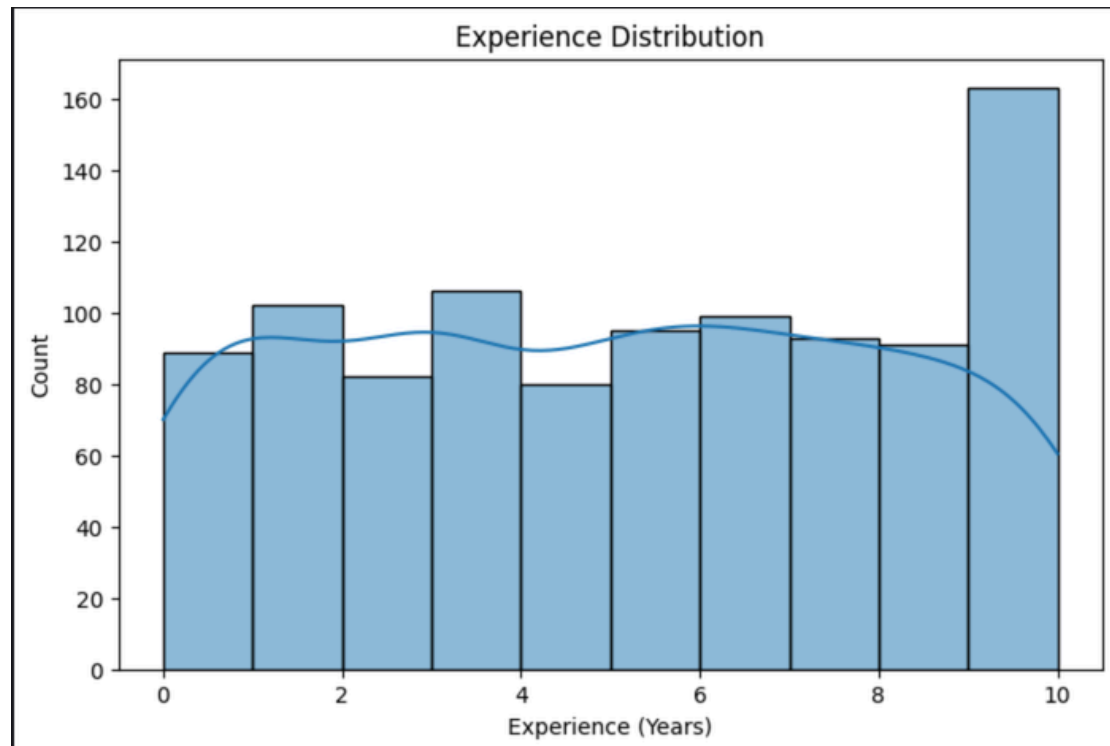
- Candidate selection is influenced by multiple attributes rather than a single factor.
- AI Score, Skills, and Experience play a strong role in predicting candidate selection.
- Candidates with better scores, more experience, and stronger profiles are more likely to be selected.
- Salary expectation can also affect the final prediction when compared with candidate profile strength.
- Feature distributions help understand the differences between selected and rejected candidates

.

Visualizations Used

- Count Plot / Bar Chart – for Selected vs Rejected candidate distribution
- Histogram – for feature distribution such as Experience, Skills, or AI Score
- Box Plot – to compare numerical features by decision
- Heatmap – to analyse correlation between features
- Feature Importance Plot – to identify the most influential variables





7. MODEL DEVELOPMENT & RESULTS

Models Applied

1. Logistic Regression

- Used as a baseline classification model.
- Provided good performance and helped establish an initial benchmark.
- After tuning, it achieved a high cross-validation score.

2. Decision Tree

- Captured non-linear relationships in the dataset effectively.
- Achieved very high accuracy on the given dataset.
- Easy to interpret but may overfit if not controlled properly.

3. Random Forest

- Delivered excellent performance and stable results.
- Reduced overfitting through ensemble learning.
- Performed strongly in predicting recruiter decisions.

Hyperparameter Tuning

Hyperparameter tuning was carried out using GridSearchCV to improve model performance and identify the best parameter combinations.

Algorithms Tuned

- Logistic Regression
- Random Forest

Best Cross Validation Scores

- Logistic Regression: 0.9975
- Random Forest: 1.00

Outcome

Hyperparameter tuning improved model reliability and confirmed that the selected models were highly effective for this dataset

Model Performance Summary

- Logistic Regression Accuracy: 0.92
- Decision Tree Accuracy: 1.00
- Random Forest Accuracy: 1.00

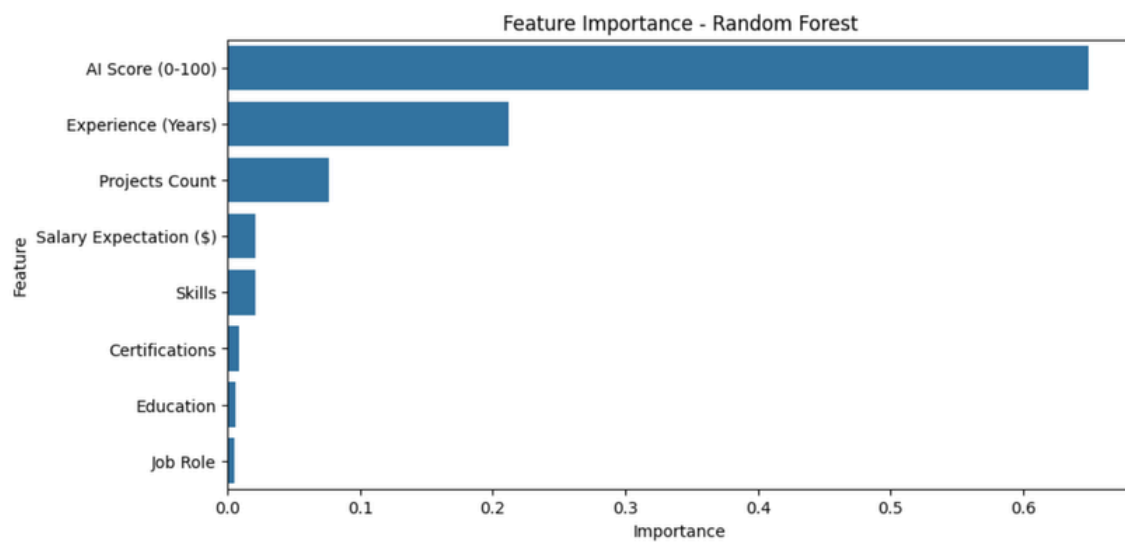
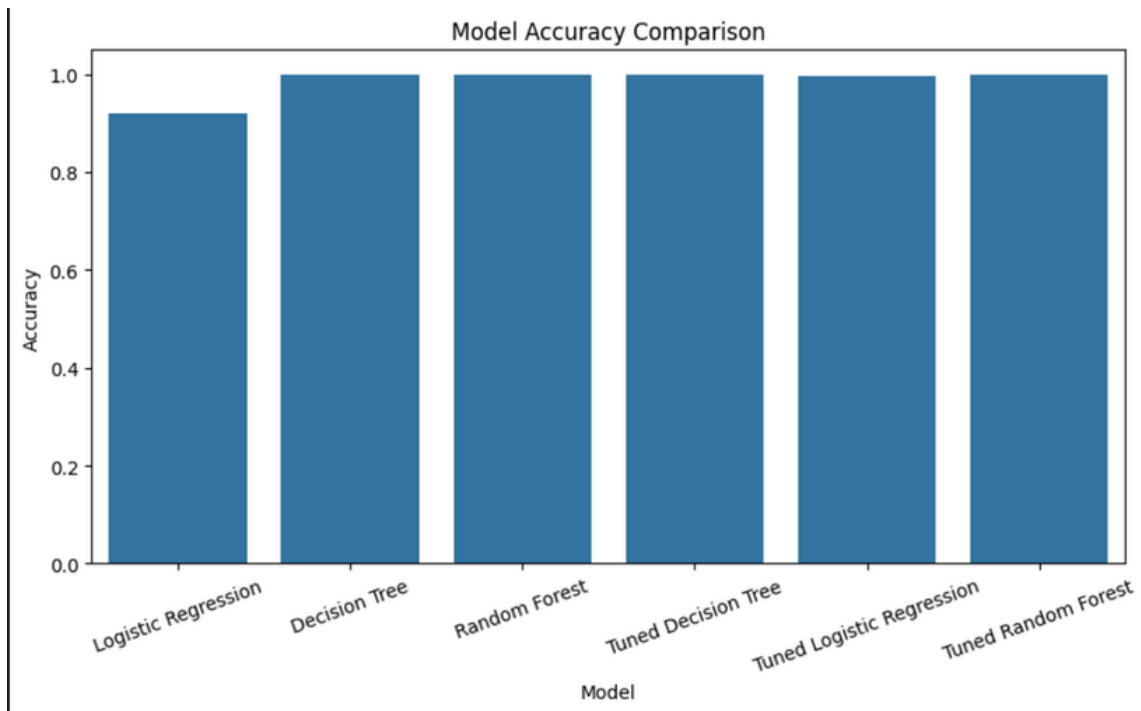
Best Model

Based on overall performance, Random Forest was considered one of the strongest models in the project because it achieved high accuracy, strong generalization, and good stability. Decision Tree also showed excellent performance on the given dataset.

Top Influential Features

From the feature importance analysis, the most influential factors in candidate screening were:

- AI Score
- Experience
- Skills
- Projects Count
- Salary Expectation



8. DASHBOARD OVERVIEW

An Excel Dashboard was created to present candidate screening insights and model outputs in a visual and structured format.

KPIs Displayed

- Total Candidates
- Selected Candidates
- Rejected Candidates
- Average AI Score
- Average Experience
- Average Salary Expectation

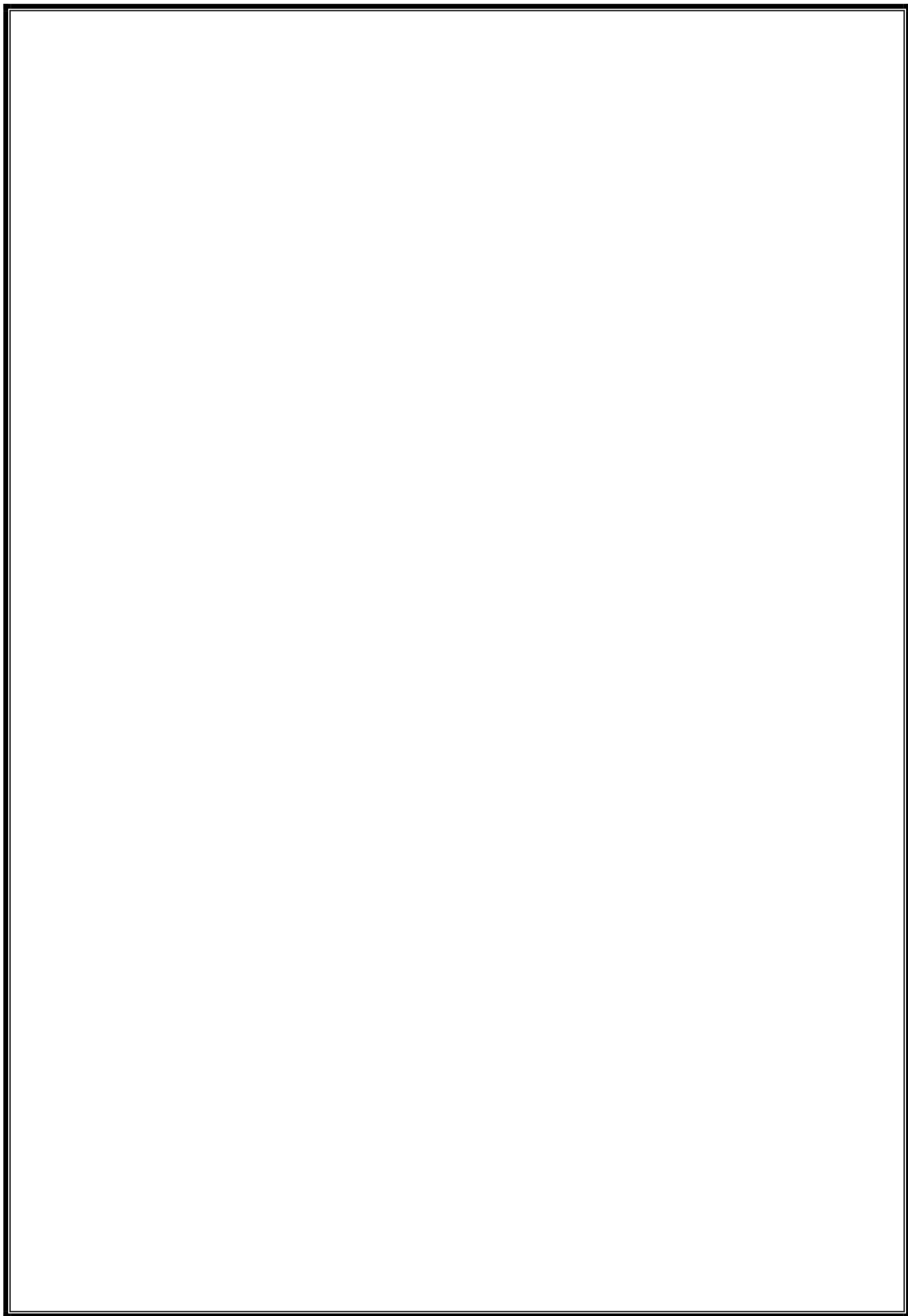
Visuals Included

- Actual Decision Distribution
- Predicted Decision Distribution
- Average AI Score by Decision
- Average Experience by Decision
- Candidate Count by Job Role
- Average Projects Count by Decision

Dashboard Purpose

The dashboard helps summarize:

- overall candidate screening statistics
- selection and rejection trends
- feature-based comparisons
- candidate profile insights for reporting and decision-making



Visuals:

- Selected vs Rejected Candidates
- Predicted Decision Distribution
- Average AI Score by Decision
- Average Experience by Decision
- Candidate Count by Job Role
- Average Projects Count by Decision

9. STREAMLIT WEB APP DEPLOYMENT

The final machine learning model was deployed using Streamlit to create an interactive web application for resume screening.

The Streamlit app allows users to enter candidate details such as:

- Skills
- Experience
- Education
- Certifications
- Job Role
- Salary Expectation
- Projects Count
- AI Score

and instantly receive a prediction as Selected or Rejected.

Key Features of the Streamlit App

- Simple and interactive user interface
- Real-time prediction
- Easy candidate input
- Practical demonstration of machine learning deployment

AI Resume Screening ML

Streamlit

+

Ask Gemini

localhost:8501

Deploy

AI-Based Resume Screening System

Skills

220

-

+

Experience (Years)

10

-

+

Education

4

-

+

Certifications

3

-

+

Job Role

2

-

+

Salary Expectation (\$)

75000

-

+

Projects Count

8

-

+

AI Resume Screening ML

Streamlit

+

Ask Gemini

localhost:8501

Deploy

4

-

+

Certifications

3

-

+

Job Role

2

-

+

Salary Expectation (\$)

75000

-

+

Projects Count

8

-

+

AI Score (0-100)

95

-

+

Predict

Prediction Value: 0

Selected

AI Resume Screening System

Skills

Experience (Years)

Education

Certifications

Job Role

Salary Expectation (\$)

Projects Count

AI Score (0-100)

Predict

Prediction Value: 0

Selected

AI Resume Screening ML

Streamlit

Ask Gemini

localhost:8501

Deploy

AI-Based Resume Screening System

Skills

5

-

+

Experience (Years)

0

-

+

Education

0

-

+

Certifications

0

-

+

Job Role

0

-

+

Salary Expectation (\$)

180000

-

+

Projects Count

0

-

+

AI Resume Screening ML

Streamlit

Ask Gemini

localhost:8501

Deploy

0

-

+

Certifications

0

-

+

Job Role

0

-

+

Salary Expectation (\$)

180000

-

+

Projects Count

0

-

+

AI Score (0-100)

10

-

+

Predict

Prediction Value: 1

Rejected

AI Resume Screening System

AI Score (0-100)

10

Rejected

10. CONCLUSION & FUTURE SCOPE

Conclusion

This project successfully demonstrates the use of machine learning classification algorithms for automating the resume screening process. By analysing candidate-related attributes such as skills, experience, education, certifications, projects, salary expectation, and AI score, the system predicts whether a candidate is likely to be selected or rejected.

The project includes data preprocessing, EDA, model building, hyperparameter tuning, feature importance analysis, Excel reporting, dashboard creation, and Streamlit deployment, making it a complete end-to-end machine learning application. The high performance of the trained models shows that machine learning can be effectively used to support faster and more structured candidate screening.

Future Scope

- Use a larger and more realistic resume dataset for stronger real-world applicability.
- Add text-based resume parsing so the system can analyse actual resume documents automatically.
- Explore advanced models such as XGBoost, LightGBM, or CatBoost for further performance improvement.
- Integrate the project with real recruitment platforms for live candidate screening.
- Add explainable AI techniques to show why a candidate was selected or rejected.
- Deploy the project as a full-scale recruitment support web application.